

# Amenable Groups Acting on Trees

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## 1 Introduction

In this note we discuss a fixed-point theorem for actions of amenable groups on trees. Recall that a group acts freely on a tree if and only if the group is free. Consequently, if a group is not free, then every action of that group on a tree has a nontrivial element fixing a point in the geometric realization of the tree. Amenability is a stronger hypothesis than non-freeness: every amenable group contains no nonabelian free subgroup, although the converse is false. Under this stronger hypothesis, one obtains a correspondingly stronger conclusion: the whole group must globally preserve a simple object naturally associated with the tree.

The goal of this note is to prove the following theorem.

**Theorem 1.1.** *Let  $\Gamma$  be an amenable group acting on a tree  $T$  by automorphisms. Then the action fixes one of the following objects:*

- (i) *a vertex;*
- (ii) *an edge, setwise, possibly with the endpoints flipped;*
- (iii) *an end;*
- (iv) *a pair of ends, setwise, possibly with the two ends flipped.*

Toward this theorem, we review some background on Boolean algebras, Stone duality, ends of trees, and the relationship between ultrafilters and orientations of trees. Not all of this material is strictly necessary for the proof of Theorem 1.1, but it motivates the compactness and orientation arguments used below.

## 2 Background on Boolean Algebras

We begin by recalling some standard definitions.

**Definition 2.1** (*Lattice, order-theoretic*). A *lattice* is a partially ordered set  $(L, \leq)$  such that every pair  $a, b \in L$  has a least upper bound, denoted  $a \vee b$ , and a greatest lower bound, denoted  $a \wedge b$ .

Equivalently, lattices can be defined algebraically.

**Definition 2.2** (*Lattice, algebraic*). A *lattice* is a set  $L$  equipped with two binary operations  $\vee$  and  $\wedge$  which are commutative and associative and satisfy the absorption laws

$$a \vee (a \wedge b) = a \quad \text{and} \quad a \wedge (a \vee b) = a$$

for all  $a, b \in L$ .

These two definitions are equivalent. The following are basic examples.

**Example 2.3.** For any set  $A$ , the power set  $\mathcal{P}(A)$ , ordered by inclusion, is a lattice. The join is union and the meet is intersection.

**Example 2.4.** The set of positive divisors of  $n$ , ordered by divisibility, is a lattice. The join is least common multiple and the meet is greatest common divisor.

**Example 2.5.** For any set  $A$ , the set of partitions of  $A$ , ordered by refinement, is a lattice. Here we use the convention that  $\alpha \leq \beta$  means that  $\alpha$  refines  $\beta$ . If  $\alpha$  and  $\beta$  are partitions of  $A$ , then  $\alpha \wedge \beta$  is the common refinement defined by

$$x \sim_{\alpha \wedge \beta} y \iff x \sim_{\alpha} y \text{ and } x \sim_{\beta} y,$$

and  $\alpha \vee \beta$  is the partition whose equivalence relation is the transitive closure of  $\sim_{\alpha} \cup \sim_{\beta}$ .

**Definition 2.6** (*Distributive lattice*). A lattice  $(L, \vee, \wedge)$  is *distributive* if

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

for all  $a, b, c \in L$ .

The lattices in Examples 2.3 and 2.4 are distributive. The partition lattice in Example 2.5 is not distributive in general.

**Definition 2.7** (*Complemented lattice*). A bounded lattice  $L$  is *complemented* if, for every  $a \in L$ , there is a  $b \in L$  such that

$$a \vee b = 1 \quad \text{and} \quad a \wedge b = 0.$$

Such an element  $b$  is called a complement of  $a$ .

**Definition 2.8** (*Boolean algebra*). A *Boolean algebra* is a complemented distributive lattice.

The lattice in Example 2.3 is a Boolean algebra, with set-theoretic complement as the Boolean complement. For Example 2.4, the positive divisors of  $n$  form a Boolean algebra if and only if  $n$  is square-free.

Every finite Boolean algebra is isomorphic to the Boolean algebra of all subsets of a finite set. More generally, as in Theorem 3.6, every Boolean algebra is isomorphic to the Boolean algebra of clopen sets in a compact, zero-dimensional, Hausdorff space.

**Definition 2.9** (*Filter and ultrafilter*). Let  $B$  be a Boolean algebra. A subset  $\mathcal{U} \subseteq B$  is a *filter* if

- (i)  $0 \notin \mathcal{U}$  and  $1 \in \mathcal{U}$ ;
- (ii) if  $a \in \mathcal{U}$  and  $a \leq b$ , then  $b \in \mathcal{U}$ ;
- (iii) if  $a, b \in \mathcal{U}$ , then  $a \wedge b \in \mathcal{U}$ .

A filter  $\mathcal{U}$  is an *ultrafilter* if it is maximal among proper filters. Equivalently,  $\mathcal{U}$  is an ultrafilter if, for every  $a \in B$ , exactly one of  $a$  and  $\neg a$  lies in  $\mathcal{U}$ .

### 3 Stone Duality

Given a Boolean algebra  $B$ , we associate to it a topological space  $S(B)$ , called its *Stone space*.

**Definition 3.1** (*Stone space*). Let  $B$  be a Boolean algebra. The *Stone space* of  $B$  is the set  $S(B)$  of ultrafilters on  $B$ , equipped with the topology generated by the basic open sets

$$N_b := \{\mathcal{U} \in S(B) : b \in \mathcal{U}\}$$

for  $b \in B$ .

Stone spaces have strong separation and compactness properties.

**Theorem 3.2.** *Stone spaces are compact, zero-dimensional, Hausdorff spaces.*

We prove this through three lemmas.

**Lemma 3.3.** *Stone spaces are zero-dimensional.*

*Proof.* Let  $B$  be a Boolean algebra. For every  $a \in B$ ,

$$S(B) \setminus N_a = N_{\neg a}.$$

Thus each  $N_a$  is clopen. Since the sets  $N_a$  form a basis,  $S(B)$  is zero-dimensional. ■

**Lemma 3.4.** *Stone spaces are Hausdorff.*

*Proof.* Let  $B$  be a Boolean algebra and let  $\mathcal{U}, \mathcal{U}'$  be distinct ultrafilters on  $B$ . Choose  $a \in B$  such that  $a \in \mathcal{U}$  and  $a \notin \mathcal{U}'$ . Since  $\mathcal{U}'$  is an ultrafilter,  $\neg a \in \mathcal{U}'$ . Therefore  $N_a$  and  $N_{\neg a}$  are disjoint open neighborhoods of  $\mathcal{U}$  and  $\mathcal{U}'$ , respectively. ■

**Lemma 3.5.** *Stone spaces are compact.*

*Proof.* Let  $B$  be a Boolean algebra. It is enough to consider covers by basic open sets. Let  $\{N_{a_i}\}_{i \in I}$  be a basic open cover of  $S(B)$ , and suppose it has no finite subcover. Then for every finite  $F \subseteq I$ ,

$$\bigcup_{i \in F} N_{a_i} \neq S(B),$$

so  $\bigcap_{i \in F} N_{\neg a_i} \neq \emptyset$ . Equivalently, the set  $\{\neg a_i : i \in I\}$  has the finite intersection property. Hence it generates a proper filter, which can be extended to an ultrafilter  $\mathcal{U}$  by Zorn's lemma. Then  $\neg a_i \in \mathcal{U}$  for every  $i \in I$ , so  $a_i \notin \mathcal{U}$  for every  $i \in I$ . Thus  $\mathcal{U} \notin \bigcup_{i \in I} N_{a_i}$ , contradicting that the family was a cover. ■

*Proof of Theorem 3.2.* This follows from Lemmas 3.3 to 3.5. ■

The converse also holds.

**Theorem 3.6** (*Stone duality*). Let  $X$  be a compact, zero-dimensional, Hausdorff space. Then there is a Boolean algebra  $B$  such that  $X$  is homeomorphic to  $S(B)$ .

*Proof.* Let  $B$  be the Boolean algebra of clopen subsets of  $X$ . Define  $\varphi : X \rightarrow S(B)$  by

$$\varphi(x) = \mathcal{U}_x := \{A \in B : x \in A\}.$$

For each  $x \in X$ ,  $\mathcal{U}_x$  is an ultrafilter.

We first show that  $\varphi$  is bijective. Let  $\mathcal{U} \in S(B)$ . Since  $\mathcal{U}$  is a proper filter, it has the finite intersection property. Since the elements of  $\mathcal{U}$  are closed and  $X$  is compact,

$$\bigcap \mathcal{U} \neq \emptyset.$$

We claim that this intersection contains at most one point. Suppose, toward a contradiction, that  $x, y \in \bigcap \mathcal{U}$  with  $x \neq y$ . Since  $X$  is Hausdorff and zero-dimensional, there is a clopen set  $C \subseteq X$  such that  $x \in C$  and  $y \notin C$ . Since  $\mathcal{U}$  is an ultrafilter, either  $C \in \mathcal{U}$  or  $X \setminus C \in \mathcal{U}$ . The first possibility contradicts  $y \in \bigcap \mathcal{U}$ , and the second contradicts  $x \in \bigcap \mathcal{U}$ . Therefore  $\bigcap \mathcal{U} = \{x\}$  for some  $x \in X$ .

It remains to see that  $\mathcal{U} = \mathcal{U}_x$ . If  $A \in \mathcal{U}$ , then  $x \in A$  because  $x \in \bigcap \mathcal{U}$ . Conversely, if  $A \in B$  and  $x \in A$ , then  $X \setminus A \notin \mathcal{U}$ , so  $A \in \mathcal{U}$  by maximality. Hence  $\mathcal{U} = \mathcal{U}_x$ .

Finally,  $\varphi$  is continuous because

$$\varphi^{-1}(N_A) = A$$

for every clopen set  $A \in B$ . Since  $X$  is compact and  $S(B)$  is Hausdorff, this continuous bijection is a homeomorphism. ■

## 4 Ends of Trees

In this section we review the necessary background on the space of ends of a tree.

**Definition 4.1.** Let  $T = (V, E)$  be a tree. A *geodesic path* in  $T$  is an injective sequence of vertices  $(v(i))_{i \in I}$ , where  $I$  is one of  $\mathbb{Z}$ ,  $\mathbb{N}$ , or  $\{0, \dots, n\}$ , such that

$$d(v(i), v(j)) = |i - j|$$

for all  $i, j \in I$ .

If  $I = \mathbb{Z}$ , the geodesic is a bi-infinite path; if  $I = \mathbb{N}$ , it is a ray; and if  $I = \{0, \dots, n\}$ , it is a finite path.

**Definition 4.2.** Let  $T = (V, E)$  be a tree, and let  $\mathcal{R}$  be the set of geodesic rays in  $T$ . Define an equivalence relation  $\sim$  on  $\mathcal{R}$  by declaring  $r \sim r'$  if and only if  $r$  and  $r'$  are eventually equal; that is, if there are  $n, n' \in \mathbb{N}$  such that

$$r(n + i) = r'(n' + i)$$

for all  $i \in \mathbb{N}$ . The *space of ends* of  $T$ , denoted  $\partial T$ , is the quotient  $\mathcal{R} / \sim$ . We write

$$\overline{T} := V \cup \partial T.$$

An end should be viewed as an infinite branch through the tree, with the starting point forgotten.

Let  $x \in V$  and  $\xi \in \partial T$ . We say that a geodesic ray  $(v(n))_{n \in \mathbb{N}}$  is *from  $x$  to  $\xi$*  if  $v(0) = x$  and  $(v(n))_{n \in \mathbb{N}}$  represents the equivalence class  $\xi$ . If  $\xi, \eta \in \partial T$ , then a bi-infinite geodesic  $(v(n))_{n \in \mathbb{Z}}$  *connects  $\xi$  to  $\eta$*  if  $(v(n))_{n \in \mathbb{N}}$  represents  $\xi$  and  $(v(-n))_{n \in \mathbb{N}}$  represents  $\eta$ . Such a geodesic is unique, up to indexing in the case where both endpoints are ends.

For  $x, y \in \bar{T}$ , let  $[x, y]$  denote the set of vertices lying on the unique geodesic from  $x$  to  $y$ . Trees have the following median property: if  $x, y, z \in \bar{T}$ , then

$$[x, y] \cap [y, z] \cap [z, x]$$

contains exactly one vertex. This vertex is called the *median* of  $x, y, z$ .

We now define the topology on  $\bar{T}$ . Given an edge  $e = xy$  of  $T$ , deleting  $e$  splits  $T$  into two connected components. We call the corresponding subsets of  $V$ , together with the ends represented by rays eventually contained in them, the two *halfspaces* determined by  $e$ .

**Definition 4.3.** The *end completion* of  $T$  is the set  $\bar{T}$  equipped with the following topology. For  $x \in \bar{T}$  and a finite set  $F \subseteq E$  of edges, let

$$U(x, F) = \{y \in \bar{T} : [x, y] \cap F = \emptyset\}.$$

The topology on  $\bar{T}$  is generated by the sets  $U(x, F)$ , where  $x \in \bar{T}$  and  $F \subseteq E$  is finite.

The set  $U(x, F)$  is the connected component of  $x$  after deleting the finitely many edges in  $F$ . Equivalently, it is the intersection of the halfspaces containing  $x$  associated with the edges in  $F$ .

The sets  $U(x, F)$  form a basis. Indeed, if

$$x \in U(x_1, F_1) \cap U(x_2, F_2),$$

then

$$x \in U(x, F_1 \cup F_2) \subseteq U(x_1, F_1) \cap U(x_2, F_2).$$

The topology on  $\bar{T}$  is Hausdorff: if  $x \neq y$ , choose an edge  $e$  on the geodesic from  $x$  to  $y$ . The two halfspaces determined by  $e$  separate  $x$  and  $y$ .

**Lemma 4.4.** *A vertex of  $T$  is isolated in  $\bar{T}$  if and only if it has finite degree.*

*Proof.* Let  $x \in V$ . If  $x$  has finite degree, let  $E_x$  be the finite set of edges incident to  $x$ . Then

$$U(x, E_x) = \{x\},$$

so  $x$  is isolated.

Conversely, suppose  $x$  is isolated. Then there is a basic open set  $U(x_0, F_0)$  such that

$$x \in U(x_0, F_0) \subseteq \{x\}.$$

If some edge  $xy$  incident to  $x$  were not in  $F_0$ , then  $y \in U(x_0, F_0)$ , a contradiction. Thus every edge incident to  $x$  lies in the finite set  $F_0$ , so  $x$  has finite degree. ■

We next discuss orientations of trees and their relationship with ultrafilters on the Boolean algebra generated by halfspaces.

**Definition 4.5.** An *orientation*  $\mathcal{O}$  of a tree  $T$  is a choice of direction for each edge. Thus, for each edge  $e = xy \in E$ , one endpoint is designated as the origin  $o(e)$  and the other as the terminus  $t(e)$ .

**Definition 4.6.** An orientation  $\mathcal{O}$  of  $T$  is *coherent* if no vertex has more than one outgoing edge.

**Lemma 4.7.** Let  $\mathcal{O}$  be a coherent orientation of  $T$ . Then all edges point toward either a single vertex or a single end of  $T$ .

*Proof.* Start at any vertex and repeatedly follow the unique outgoing edge, if one exists. Since  $T$  has no cycles, this process either terminates at a vertex with no outgoing edge or determines a ray, hence an end.

If the process terminates, the terminal vertex is independent of the starting point: two distinct terminal vertices would force the path between them to contain a vertex with two outgoing edges. Similarly, if the process determines an end, the resulting end is independent of the starting point; otherwise the geodesic connecting two resulting rays would again contain a vertex with two outgoing edges. Therefore the whole orientation points toward a unique vertex or a unique end. ■

Let  $\mathcal{B}(T)$  denote the Boolean algebra generated by the halfspaces of  $T$ . Equivalently,  $\mathcal{B}(T)$  is the Boolean algebra of all subsets  $A \subseteq V$  whose edge boundary

$$\delta A := \{xy \in E : x \in A \text{ and } y \notin A\}$$

is finite.

If  $A \in \mathcal{B}(T)$ , define  $\hat{A} \subseteq \bar{T}$  by

$$\hat{A} := A \cup \{\xi \in \partial T : \text{some/every ray representing } \xi \text{ is eventually contained in } A\}.$$

This is well-defined because every ray crosses the finite set  $\delta A$  only finitely many times.

**Lemma 4.8.** Let  $T$  be a tree. For every ultrafilter  $\mathcal{U}$  on  $\mathcal{B}(T)$ , there is a coherent orientation  $\mathcal{O}_{\mathcal{U}}$  of  $T$  such that exactly one of the following holds:

- (i)  $\mathcal{O}_{\mathcal{U}}$  points toward a vertex  $v$ , and  $\mathcal{U} = \{A \in \mathcal{B}(T) : v \in A\}$ ;
- (ii)  $\mathcal{O}_{\mathcal{U}}$  points toward an end  $\xi$ , and  $\mathcal{U} = \{A \in \mathcal{B}(T) : \xi \in \hat{A}\}$ .

*Proof.* Let  $\mathcal{U}$  be an ultrafilter on  $\mathcal{B}(T)$ . For each edge  $xy$ , let  $H(x, y)$  be the halfspace consisting of the vertices and ends lying in the component of  $T \setminus \{xy\}$  containing  $y$ . Exactly one of  $H(x, y) \cap V$  and  $H(y, x) \cap V$  lies in  $\mathcal{U}$ . Orient the edge  $xy$  toward  $y$  if and only if  $H(x, y) \cap V \in \mathcal{U}$ .

This orientation is coherent. Indeed, if two distinct edges  $xy$  and  $xz$  were both oriented away from  $x$ , then the halfspaces  $H(x, y) \cap V$  and  $H(x, z) \cap V$  would be disjoint elements of  $\mathcal{U}$ , contradicting closure of  $\mathcal{U}$  under finite meets and properness.

By Lemma 4.7, the orientation points toward a vertex  $v$  or an end  $\xi$ . If it points toward  $v$ , then, for every edge  $xy$ , the halfspace selected by  $\mathcal{U}$  is precisely the one containing  $v$ . Since the halfspaces generate  $\mathcal{B}(T)$ , it follows that

$$\mathcal{U} = \{A \in \mathcal{B}(T) : v \in A\}.$$

The proof in the end case is the same: for every edge  $xy$ , the selected halfspace is precisely the one containing  $\xi$ , and hence

$$\mathcal{U} = \{A \in \mathcal{B}(T) : \xi \in \hat{A}\}.$$

■

**Theorem 4.9.** *Let  $T$  be a tree. Then the Stone space  $S(\mathcal{B}(T))$  is homeomorphic to the end completion  $\overline{T}$ .*

*Proof.* Define

$$\Phi : \overline{T} \rightarrow S(\mathcal{B}(T))$$

by

$$\Phi(x) = \mathcal{U}_x := \{A \in \mathcal{B}(T) : x \in \hat{A}\},$$

where for a vertex  $x \in V$  the condition  $x \in \hat{A}$  is simply  $x \in A$ .

The set  $\mathcal{U}_x$  is an ultrafilter for every  $x \in \overline{T}$ . Conversely, Lemma 4.8 shows that every ultrafilter on  $\mathcal{B}(T)$  is of the form  $\mathcal{U}_x$  for a unique  $x \in \overline{T}$ . Thus  $\Phi$  is bijective.

For each  $A \in \mathcal{B}(T)$ ,

$$\Phi^{-1}(N_A) = \hat{A}.$$

Since  $\delta A$  is finite,  $\hat{A}$  is a finite union of basic sets of the form  $U(x, \delta A)$ , and is therefore clopen in  $\overline{T}$ . Thus  $\Phi$  is continuous. Conversely, the same identity shows that  $\Phi(\hat{A}) = N_A$ , so  $\Phi$  sends the clopen basis generated by finite-boundary sets to the clopen basis of  $S(\mathcal{B}(T))$ . Hence  $\Phi$  is a homeomorphism. ■

**Corollary 4.10.** *Let  $\Gamma$  act by automorphisms on a tree  $T$ . Then this action extends uniquely to a continuous action  $\Gamma \curvearrowright \overline{T}$  on the end completion of  $T$ .*

*Proof.* Each automorphism of  $T$  sends halfspaces to halfspaces and therefore induces an automorphism of the Boolean algebra  $\mathcal{B}(T)$ . By functoriality of Stone duality, it induces a homeomorphism of  $S(\mathcal{B}(T))$ . Using the homeomorphism from Theorem 4.9, this gives a continuous action on  $\overline{T}$  extending the original action on  $V$ . ■

## 5 Proof of the Main Theorem

We now prove the theorem.

**Theorem 1.1.** *Let  $\Gamma$  be an amenable group acting on a tree  $T$  by automorphisms. Then the action fixes one of the following objects:*

- (i) a vertex;
- (ii) an edge, setwise, possibly with the endpoints flipped;
- (iii) an end;
- (iv) a pair of ends, setwise, possibly with the two ends flipped.

*Proof.* Let  $\Gamma \curvearrowright T$  be an action by automorphisms. By Corollary 4.10, this action extends to a continuous action on the compact space  $\overline{T}$ . Since  $\Gamma$  is amenable, every continuous action of

$\Gamma$  on a compact Hausdorff space admits an invariant Borel probability measure. Let  $\mu$  be a  $\Gamma$ -invariant Borel probability measure on  $\overline{T}$ .

For adjacent vertices  $x, y \in V$ , let  $H(x, y) \subseteq \overline{T}$  denote the halfspace consisting of the vertices and ends lying in the component of  $T \setminus \{xy\}$  containing  $y$ . Thus  $H(x, y)$  and  $H(y, x)$  are complementary clopen subsets of  $\overline{T}$ .

Let  $A$  be the set of unoriented edges  $\{x, y\}$  such that

$$\mu(H(x, y)) = \mu(H(y, x)) = \frac{1}{2}.$$

This condition is independent of the orientation of the edge, and  $A$  is invariant under the action of  $\Gamma$ .

We first observe that  $A$  is connected whenever it is nonempty. Suppose  $x_0, \dots, x_n$  is a geodesic path in  $T$  and that the first and last edges,  $\{x_0, x_1\}$  and  $\{x_{n-1}, x_n\}$ , lie in  $A$ . Then

$$H(x_{n-1}, x_n) \subseteq H(x_i, x_{i+1}) \subseteq H(x_0, x_1)$$

for every  $0 \leq i < n$ . Since the outer two halfspaces have measure  $1/2$ , each intermediate halfspace also has measure  $1/2$ . Hence every edge on the path lies in  $A$ .

Next, every vertex in the subgraph induced by  $A$  has degree at most 2. Indeed, suppose for contradiction that  $x_0, x_1, x_2$  are three distinct neighbors of a vertex  $y$  and that the edges  $\{x_i, y\}$  all lie in  $A$ . Then the three halfspaces

$$H(y, x_0), \quad H(y, x_1), \quad H(y, x_2)$$

are pairwise disjoint, and each has measure  $1/2$ , which is impossible.

Thus, if  $A$  is nonempty, it is a connected graph of maximum degree at most 2. There are three possibilities.

First, suppose  $A$  is finite. Then  $A$  is a finite path. The action of  $\Gamma$  preserves this path, and therefore fixes its barycenter. If the path has even length, this barycenter is a vertex; if it has odd length, it is an edge, fixed setwise.

Second, suppose  $A$  is an infinite ray. Then its unique endpoint is fixed by every automorphism preserving  $A$ .

Third, suppose  $A$  is a bi-infinite line. Then the two ends represented by this line are preserved setwise by  $\Gamma$ .

It remains to consider the case  $A = \emptyset$ . In this case, for every edge  $xy$ , exactly one of  $H(x, y)$  and  $H(y, x)$  has measure strictly greater than  $1/2$ . Orient the edge toward the halfspace of larger measure. This orientation is  $\Gamma$ -invariant by invariance of  $\mu$ .

The orientation is coherent: if two distinct edges  $xy$  and  $xz$  were both oriented away from  $x$ , then the disjoint halfspaces  $H(x, y)$  and  $H(x, z)$  would each have measure greater than  $1/2$ , a contradiction. By Lemma 4.7, this coherent orientation points toward a vertex or an end. Since the orientation is  $\Gamma$ -invariant, that vertex or end is fixed by the action of  $\Gamma$ . ■